High Performance Computing

Report Mandelbrot Algorithm

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Contents

1 Architectures

For this project I have used three workstation, my personal one, the workstation provided by the University and Google Colab.

1.1 Personal Workstation

HP Envy 17-ae103nl

CPU	
Model	Intel Core i7-6500U @ 2.50 GHz
Architecture	x86_64
Physical cores	2
Logical threads	4 (Hyper-Threading)

GPU	
Model	NVIDIA GeForce GTX 950M
CUDA Compute Capability	5.0
Driver version	572.16
CUDA Version	12.8

Software and Setup	
Visual Studio	Visual Studio Community 2022 (C++ Desktop workload)
CUDA Toolkit	Installed manually
WSL2	Enabled
NVIDIA Drivers	Version 572.16 (for WSL2)

1.2 University Workstation

Room 210

CPU	
Model	Intel Core i9-12900K (12th Gen)
Architecture	x86_64
CPU(s)	24
Threads per core	2
Cores per socket	16
Socket(s)	1
Max frequency	4.02 GHz
Profiling tools	Intel VTune Profiler, Intel Advisor

To use the Intel oneAPI tools, the environment was loaded with:

```
source /opt/intel/oneapi/setvars.sh
```

1.3 Google Colab

CPU	
Model	Intel(R) Xeon(R) CPU @ 2.20GHz
Architecture	x86_64
Physical cores	1
Logical threads	2 (Hyper-Threading)
Socket(s)	1

GPU	
Model	NVIDIA Tesla T4
CUDA Compute Capability	7.5
Driver version	550.54.15
CUDA Version	12.8

2 Mandelbrot Set

2.1 Definition

The Mandelbrot fractal is defined as a set of points in the complex plane. Each point corresponds to a complex number c.

For every c, we consider the iterative sequence:

$$z_0 = 0, \qquad z_{n+1} = z_n^2 + c$$

We then observe the behavior of z_n as $n \to \infty$.

2.1.1 Point that belongs to the Mandelbrot set

Take $c = -1$. Then the sequence is

$$z_0 = 0, \quad z_1 = 0^2 + (-1) = -1, \quad z_2 = (-1)^2 + (-1) = 0, \quad z_3 = 0^2 + (-1) = -1, \ldots$$

The results remain bounded, so $c = -1$ belongs to the Mandelbrot set.

2.1.2 Point that does NOT belong to the Mandelbrot set

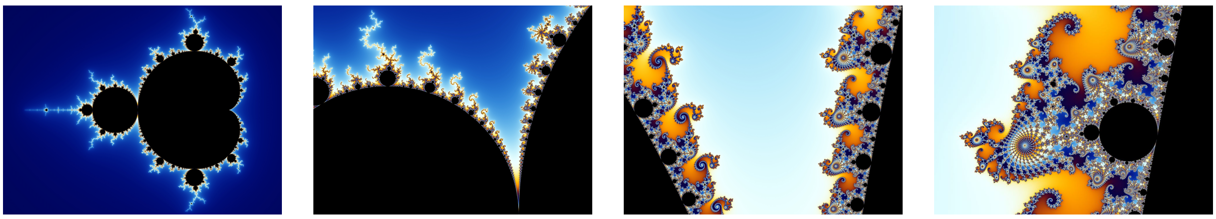
Take $c = 1$. Then the sequence is

$$z_0 = 0, \quad z_1 = 0^2 + 1 = 1, \quad z_2 = 1^2 + 1 = 2, \quad z_3 = 2^2 + 1 = 5, \ldots$$

The values grow up to infinite, so $c = 1$ does not belong to the Mandelbrot set.

2.2 Images

This process gives rise to the famously intricate structure of the Mandelbrot fractal. Here a simple explanation [1] or here a more complex one [2] and here an online demo [3].



The color represents how quickly the values of z_n grow:

Black - the sequence remains bounded;

Other colors - the sequence escapes to infinity, with the color indicating how fast it diverges.

3 Conclusions

References

[1] Simple Explanation of the Mandelbrot Set.
https://www.wikihow.com/Plot-the-Mandelbrot-Set-By-Hand

[2] Mandelbrot Set – Formal Explanation.
https://en.wikipedia.org/wiki/Mandelbrot_set

[3] Online Mandelbrot Demo.
https://math.hws.edu/eck/js/mandelbrot/MB.html